

Mohamed Youssef

Brooklyn, NY | mohamed.youssef@nyu.edu | (917) 909-9858 | [LinkedIn](#)

EDUCATION

New York University Tandon School of Engineering

Brooklyn, NY

Bachelor of Science in Electrical and Computer Engineering

Class of 2028

- GPA 3.6/4.0; Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Intro to Engineering and Design, Intro to Electrical and Computer Engineering, Intro to Circuits
- Tandon Honors Program: [Global Leaders and Scholars in STEM](#), a prestigious cohort of 25 students chosen for academic excellence, leadership, and research potential.
- Clubs & Programs: HEOP, TRIO Scholars, SHPE, MSA

LEADERSHIP EXPERIENCE

NYU MakerSpace Teaching Assistant

Brooklyn, NY

Training Team Member/LPKF Specialist

Jan 2025 - Present

- Serve as one of two LPKF PCB milling specialist, training engineering students and peers on circuit board fabrication workflows, including Gerber file preparation, substrate setup, and machine calibration
- Guide students in operating 3+ advanced prototyping machines while fostering a collaborative and inclusive learning environment
- Design and led a Raspberry Pi workshop, teaching 10+ students introductory programming and hardware skills
- Maintain lab safety and accessibility by enforcing MakerSpace protocols and assisting 15+ students daily in applying engineering design and manufacturing skills
- Collaborate directly with the manager and a 3-member training team to develop detailed machine operation and safety guides, improving onboarding efficiency for new users

YMCA of Greater New York

Queens, NY

Lifeguard/Swim Instructor

Jun 2022 - Sept 2025

- Empower over 50 students of all ages and backgrounds by tailoring swim lessons to individual needs, ensuring an inclusive learning experience
- Lead 2+ weekly swimming classes, fostering skill development in a diverse group of students
- Maintain vigilant surveillance of the pool, resulting in a reduction of incidents for patrons
- Certified in administering first-aid and treating open wounds, proficient in CPR and in the use of an AED

PROJECTS

Wireless 6G

Brooklyn, NY

RF Simulation Team/Manufacturing Team

Jan 2025 - Present

- Design and simulate high-frequency microstrip transmission lines using Ansys HFSS and Keysight ADS, modeling electromagnetic behavior and verifying performance through S-parameter analysis
- Create and parameterize thru connectors and calibration standards in HFSS, applying variable-driven design to streamline iterative testing
- Transition simulation results to physical prototypes via LPKF PCB milling, troubleshooting Gerber file export and layout generation workflows
- Compare simulated vs. measured data to refine material parameters and improve accuracy of RF behavior modeling

Motion-Detecting Scarecrow

Brooklyn, NY

Programmer/Arduino

Sep 2024 - Dec 2024

- Collaborate with a 4-member team to develop a functional prototype, delegating tasks efficiently
- Manage a \$100 budget to assemble a cost-effective prototype using 3D-printed and off-the-shelf parts
- Design and 3D-print a CAD enclosure to integrate electrical components seamlessly
- Program a motion-detecting scarecrow to deter unwanted animals in a garden
- Incorporate diverse team feedback to iteratively refine prototype functionality

ADDITIONAL INFORMATION

Languages: Fluent in Arabic, basic conversational skills in Spanish

Skills: Python, C++, HFSS, Advanced Design System, Arduino, Raspberry Pi, Fusion 360

Machinery: PCB Milling, 3D Printers, Laser Cutters